

Derivative Norm Spectra and Dual Moment Geometries of the Fabius–Rvachev System

**Thue–Morse equimeasurability, inverse-Fabius beta transforms,
Gamma–Bell operators, and shifted Stieltjes–Wigert energy measures**

Research report prepared for Vladimir Reshetnikov’s ProveIt development

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Abstract

Let F be the bounded Fabius distribution function, $G = F^{-1}$ its inverse on $[0, 1]$, $\text{up}(x) = 1 - F(|x|)$ on $[-1, 1]$, and

$$\Phi(\xi) = \widehat{\text{up}}(\xi) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi\xi}{2^j}\right)$$

under the Fourier convention $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx$. The recursive documentation tree of the ProveIt Fabius development was audited at commit 0341bbf91cb867f4510bc8431cf59878208de771. Existing reports already contain the dyadic derivative tiling, integer-zero arithmetic, sharp Fourier-decay regimes, endpoint Lambert expansions, q -binomial acceleration, inverse-quantile reversion, and several spectral determinant constructions. Those results are treated as baseline rather than renamed.

The first new package extracted here begins with an exact equimeasurability theorem. If $T_n = n(n+1)/2$, then $2^{-T_n} |\text{up}^{(n)}|$ has exactly the same value distribution on $[-1, 1]$ as up . The signs on the 2^n cells are the length- 2^n Thue–Morse block. Consequently every rearrangement-invariant norm, including every L^p quasi-norm $0 < p \leq \infty$, obeys

$$\left\| \text{up}^{(n)} \right\|_p = 2^{T_n} \|\text{up}\|_p.$$

The level sets are also exact:

$$\left| \{x : |\text{up}^{(n)}(x)| > 2^{T_n} t\} \right| = 2G(1-t), \quad 0 < t < 1.$$

Thus the complete inverse Fabius function is encoded in the value statistics of every nonzero derivative.

Writing $A_p = \frac{1}{2} \|\text{up}\|_p^p$, one obtains the exact beta transform

$$A_p = p \int_0^1 (1-y)^{p-1} G(y) dy.$$

For $\mathcal{D} = y \, d/dy$, the dilation variable $Z_p = pY_p$ with $Y_p \sim \text{Beta}(1, p)$ has Mellin symbol

$$\mathbb{E} Z_p^s = p^s \frac{\Gamma(p+1)\Gamma(1+s)}{\Gamma(p+1+s)}.$$

This yields a new Euler–Gamma differential calculus

$$A_p \sim G - \gamma \mathcal{D}G + \frac{\gamma^2 + \pi^2/6}{2} \mathcal{D}^2 G - \frac{\gamma^3 + \gamma\pi^2/2 + 2\zeta(3)}{6} \mathcal{D}^3 G + \dots \quad \text{at } y = p^{-1}.$$

The coefficients are complete Bell polynomials in log-exponential cumulants; finite- p corrections are generated by Bernoulli polynomials through the gamma-ratio expansion. Combining this operator with the periodic inverse-Fabius endpoint expansion gives

$$\|\text{up}\|_p^p = 2G(p^{-1}) \left(1 - \frac{\gamma}{\rho} + \mathcal{O}(\rho^{-2}) \right), \quad \rho = \sqrt{\frac{2 \log p}{\log 2}},$$

and, more sharply, shows that Euler's constant subtracts from the first periodic inverse correction. In particular

$$1 - \|\text{up}\|_p \sim \frac{\sqrt{2 \log 2 \log p}}{p}.$$

A phase-locked sequence of p -norms recovers the nonconstant Gamma-zeta endpoint fluctuation itself.

Parseval turns the same norm law into a second, frequency-side geometry. The pushforward of $|\Phi(\xi)|^2 d\xi / \|\text{up}\|_2^2$ by $z = (2\pi\xi)^2$ has moments

$$m_n = 2^{n(n+1)} = q^{-n(n+1)/2}, \quad q = \frac{1}{4}.$$

It is therefore an explicit nonclassical representing measure for a shifted Stieltjes–Wigert moment problem. A lognormal measure has the same moments but a different density, proving indeterminacy directly. We calculate the Hankel determinants, monic orthogonal polynomials, recurrence coefficients, and Gaussian- q -binomial formulae, and derive the spectral q -Pearson law

$$w_{\text{sp}}(4z) = \frac{1}{2} \text{sinc}^2(\sqrt{z}) w_{\text{sp}}(z).$$

The report closes with numerical checks, explicit conjectures, a phase-locked confluent q -Richardson program, and a Lean formalization roadmap. Claims labelled “new” are new relative to the pinned repository corpus after a formula- and terminology-level collision audit; no universal publication-priority claim is made.

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1 Scope, corpus audit, and novelty boundary

1.1 What was inspected

The recursive tree under <https://github.com/VladimirReshetnikov/ProveIt/tree/0341bbf91cb867f4510bc8431cf59878208de771/Analysis/FabiusFunction/docs> contained 68 *.tex artifacts at the pinned commit. Every path was included in the audit. The living expositions, active frontier reports, Fourier-decay chain, imported paper transcriptions, and mathematically distinctive archival deltas were read. The archive deliberately preserves first versions and provenance; its own ledger explains that byte-identical or superseded descendants should not be treated as independent contemporary claims. Accordingly, archived variants were checked for distinctive content but deduplicated when building the novelty map. The complete path ledger appears in [Appendix A](#).

The most important collision checks were against the following current reports:

- the primary Fabius–Rvachev exposition and mathematical glossary;
- the consolidated non-formalized frontier volume;
- the inverse-frontier report, including periodic Lambert–Bell quantile reversion;
- the complete Fourier-decay synthesis and its comparative audits;
- the finite-block/q-Richardson/deconvolution report;
- the half-integer, arithmetic-ray, spectral determinant, Pascal–Rvachev, and Gamma-duality frontier reports;
- all archived q-calculus, dyadic-asymptotic, Thue–Morse, and repeated-integration studies; and
- the seven imported paper sources.

Targeted searches were then run for the exact concepts and formula shapes used below: “equimeasurable derivative”, “all- L^p derivative norm”, “Beta(1, p) inverse Fabius”, “ $\Gamma(1 + \mathscr{D})$ ”, “Stieltjes–Wigert”, “Denjoy–Carleman”, and the exact moment sequence $2^{n(n+1)}$. No collision was found for the theorem package developed here. The underlying derivative tiling itself is present in the corpus and is explicitly marked as baseline.

1.2 Status convention

Three proof-status labels. *Corpus baseline* means a statement already present, perhaps in another normalization, in the inspected repository or one of its imported papers. *New deduction* means a statement and proof not located in that corpus after the collision audit. *Conjecture* means a precise, numerically or structurally motivated statement for which this report does not claim a proof. None of these labels asserts global publication priority.

The following are corpus baseline and are not claimed as new:

- (i) the identities relating F , up, and the signed global extension;
- (ii) the differential-refinement law for up and its finite Thue–Morse derivative tiling;
- (iii) the infinite sinc product and its integer zero multiplicities;
- (iv) the exact dyadic values of F , their q -Pochhammer and Gaussian-binomial forms;
- (v) the corrected endpoint expansion with a nonconstant periodic Gamma–zeta term;
- (vi) the inverse endpoint expansion in the lower-Lambert coordinate, including Bell reversion and quantile elasticity;

- (vii) the distinct pointwise, metric, L^1 , L^2 , and extremal Fourier-decay exponents;
- (viii) the existing finite-block, q-Richardson, convolution-tail, spectral determinant, and Pascal–Rvachev constructions.

New deductions proved in this report. The main new results are: exact derivative equimeasurability; scaling in every rearrangement-invariant norm; exact inverse-Fabius level-set laws; a Hausdorff reconstruction of F from integer derivative norms; the inverse-Fabius beta transform; an exact Mellin symbol and asymptotic Euler–Gamma/Bell/Bernoulli differential calculus; a large- p Lambert law with an Euler-constant shift and phase tomography; a shifted Stieltjes–Wigert spectral-energy measure; explicit Hankel determinants and q-binomial monic orthogonal polynomials; a nonclassical sinc-square q-Pearson equation; and the exact Denjoy–Carleman weight and associated periodic Legendre envelope generated by the derivative norms.

Post-audit Lean status: a completed derivative-distribution core. The current source file `Analysis/FabiusFunction/Lean/FabiusFunction/RvachevDerivativeDistribution.lean` now formalizes the exact dyadic cell coordinate and tiling, every signed cell-midpoint value, the all-order signed finite-mixture identity for continuous tests with values in an arbitrary real Banach space, the normalized absolute continuous-test identity, the equality of the two absolute Borel pushforward measures on Lebesgue measure restricted to $[-1, 1]$, and every nonnegative real absolute moment. This is a post-audit proof-status update; it does not alter the pinned novelty comparison above. The report as a whole is not thereby formalized. In particular, the simplified signed Rademacher law as a Borel pushforward, the `eLpNorm` and general rearrangement-invariant norm ladder, the explicit inverse-Fabius level-set laws, the beta transform and its asymptotics, and all spectral-moment, Hankel, q-Pearson, and q-binomial layers below remain without matching declarations.

2 Normalization and established structural identities

Let $F : \mathbb{R} \rightarrow [0, 1]$ be the bounded Fabius function, with $F = 0$ on $(-\infty, 0]$, $F = 1$ on $[1, \infty)$, symmetry

$$F(1 - x) = 1 - F(x), \quad 0 \leq x \leq 1, \quad (2.1)$$

and differential law $F'(x) = 2F(2x)$ on $0 < x < 1/2$, continued by symmetry. Let

$$\text{up}(x) = \begin{cases} 1 - F(|x|), & |x| \leq 1, \\ 0, & |x| \geq 1. \end{cases} \quad (2.2)$$

Then up is even, nonnegative, C^∞ , supported on $[-1, 1]$, satisfies $\text{up}(0) = 1$ and $\int \text{up} = 1$, and obeys

$$\text{up}'(x) = 2(\text{up}(2x + 1) - \text{up}(2x - 1)). \quad (2.3)$$

Its Fourier transform is

$$\Phi(\xi) := \widehat{\text{up}}(\xi) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi \xi}{2^j}\right), \quad \text{sinc } z = \frac{\sin z}{z}. \quad (2.4)$$

Write

$$\varepsilon_k := (-1)^{s_2(k)}, \quad T_n := \binom{n+1}{2} = \frac{n(n+1)}{2}, \quad (2.5)$$

where $s_2(k)$ is the binary digit sum. The identities $\varepsilon_{2k} = \varepsilon_k$ and $\varepsilon_{2k+1} = -\varepsilon_k$ will be used repeatedly.

Theorem 2.1 (Corpus baseline: exact derivative tiling). *For every $n \geq 0$ and $x \in \mathbb{R}$,*

$$\boxed{\text{up}^{(n)}(x) = 2^{T_n} \sum_{k=0}^{2^n-1} \varepsilon_k \text{up}(2^n x + 2^n - 1 - 2k).} \quad (2.6)$$

Equivalently, for $0 \leq k < 2^n$ and $-1 \leq y \leq 1$,

$$\boxed{\text{up}^{(n)}\left(-1 + \frac{2k+1+y}{2^n}\right) = 2^{T_n} \varepsilon_k \text{up}(y).} \quad (2.7)$$

Proof. The case $n = 0$ is immediate. Differentiate (2.6). Each derivative of an inner copy contributes 2^n , and (2.3) contributes another factor 2. The positive branch has index $2k$ and the negative branch index $2k+1$; the Thue–Morse identities above give exactly the signs ε_{2k} and ε_{2k+1} . The scalar grows from 2^{T_n} to $2^{T_n+n+1} = 2^{T_{n+1}}$.

For the cell form, the support condition $|2^n x + 2^n - 1 - 2j| \leq 1$ partitions $[-1, 1]$ into the 2^n closed cells

$$I_{n,k} = \left[-1 + \frac{2k}{2^n}, -1 + \frac{2k+2}{2^n}\right].$$

On the interior of $I_{n,k}$ only the k th summand survives. At common boundaries every potentially overlapping copy is evaluated at ± 1 and hence vanishes to infinite order. Solving for the local coordinate gives (2.7). $\square$

Remark 2.2. The corpus already exploits this tiling for derivative values, zeros, and discrete Thue–Morse structure. The new step is to view (2.7) as a measure-preserving value-distribution identity and push that observation through normed function spaces, inverse quantiles, Mellin operators, and spectral moments.

Exact Lean parity for the dyadic cells. The current definition `rvachevDerivativeCell` is exactly

$$c_{n,k}(y) := -1 + \frac{2k+1+y}{2^n}.$$

It is total in n and k . For $k < 2^n$ it maps $[-1, 1]$ into $[-1, 1]$; the first left endpoint is -1 , adjacent endpoints agree, and the left endpoint after all 2^n cells is 1 . These facts are `rvachevDerivativeCell_mem_Icc`, `rvachevDerivativeCell_zero_neg_one`, `rvachevDerivativeCell_one_eq_succ_neg_one`, and `rvachevDerivativeCell_two_pow_neg_one`; the affine dilation about the midpoint is `rvachevDerivativeCell_eq_div_add`.

The signed and absolute formulas (2.7) are `iteratedDeriv_rvachev_cell` and `abs_iteratedDeriv_rvachev_cell`. They include $n = 0$, every valid cell, and both cell endpoints. The midpoint specialization

$$\text{up}^{(n)}\left(-1 + \frac{2k+1}{2^n}\right) = 2^{T_n} \varepsilon_k$$

is the separate theorem `iteratedDeriv_rvachev_cell_zero`. All declarations in this box are in `RvachevDerivativeDistribution.lean`.

3 Exact equimeasurability and the complete derivative norm spectrum

3.1 Absolute and signed value distributions

Let U be uniform on $[-1, 1]$ and let R be an independent Rademacher variable, $\mathbb{P}(R = 1) = \mathbb{P}(R = -1) = 1/2$.

Theorem 3.1 (Derivative equimeasurability). *For every $n \geq 0$,*

$$\boxed{2^{-T_n} \left| \text{up}^{(n)}(U) \right| \stackrel{\text{d}}{=} \text{up}(U)}. \quad (3.1)$$

For every $n \geq 1$,

$$\boxed{2^{-T_n} \text{up}^{(n)}(U) \stackrel{\text{d}}{=} R \text{up}(U)}. \quad (3.2)$$

Equivalently, for every Borel function H for which the integrals exist,

$$\frac{1}{2} \int_{-1}^1 H\left(2^{-T_n} \left| \text{up}^{(n)}(x) \right| \right) dx = \frac{1}{2} \int_{-1}^1 H(\text{up}(y)) dy, \quad (3.3)$$

$$\frac{1}{2} \int_{-1}^1 H\left(2^{-T_n} \text{up}^{(n)}(x)\right) dx = \frac{1}{4} \int_{-1}^1 (H(\text{up}(y)) + H(-\text{up}(y))) dy \quad (n \geq 1). \quad (3.4)$$

Proof. Integrate cell by cell and use (2.7). The affine map from each cell to $[-1, 1]$ has Jacobian 2^{-n} , hence

$$\int_{I_{n,k}} H\left(2^{-T_n} \left| \text{up}^{(n)}(x) \right| \right) dx = 2^{-n} \int_{-1}^1 H(\text{up}(y)) dy.$$

Summing 2^n equal terms proves (3.3). For $n \geq 1$, exactly half of the length- 2^n Thue–Morse block has sign $+1$ and half sign -1 , proving (3.4). $\square$

Exact Lean parity for the distribution identities. The signed theorem presently exposed by Lean is the following all-order finite mixture, not yet its Rademacher simplification. If E is any real Banach space and $H : \mathbb{R} \rightarrow E$ is continuous, then for every $n \geq 0$,

$$\int_{-1}^1 H\left(\text{up}^{(n)}(x)\right) dx = 2^{-n} \sum_{k=0}^{2^n-1} \int_{-1}^1 H(2^{T_n} \varepsilon_k \text{up}(y)) dy. \quad (3.5)$$

This is `intervalIntegral_comp_iteratedDeriv_rvachev`. In particular, it includes the order-zero boundary without introducing a fictitious independent sign.

The same Banach-valued continuous-test API proves

$$\int_{-1}^1 H\left(\left| \text{up}^{(n)}(x) \right| \right) dx = \int_{-1}^1 H(2^{T_n} \text{up}(y)) dy, \quad (3.6)$$

$$\int_{-1}^1 H\left(2^{-T_n} \left| \text{up}^{(n)}(x) \right| \right) dx = \int_{-1}^1 H(\text{up}(y)) dy. \quad (3.7)$$

These are `intervalIntegral_comp_abs_iteratedDeriv_rvachev` and `intervalIntegral_comp_normalized_abs_iteratedDeriv_rvachev`.

More strongly on the absolute side, let λ_I be Lebesgue measure restricted to $I = [-1, 1]$, and write $f_{\#} \mu$ for pushforward. Lean proves the literal Borel-measure equality

$$\left(x \mapsto 2^{-T_n} \left| \text{up}^{(n)}(x) \right| \right)_{\#} \lambda_I = \text{up}_{\#} \lambda_I. \quad (3.8)$$

Its declaration is `map_normalized_abs_iteratedDeriv_rvachev_restrict_Icc`. What is not yet a named theorem is the signed Borel pushforward simplified, for $n \geq 1$, to $R \text{up}(U)$. Thus (3.2) and the full Borel formulation of (3.4) remain on the frontier even though (3.5) is their exact continuous-test finite-cell precursor.

Corollary 3.2 (All absolute moments and signed moments). *For every real $p \geq 0$, interpreting the powers as real powers and using $0^0 = 1$ at $p = 0$,*

$$\int_{-1}^1 |\text{up}^{(n)}(x)|^p dx = (2^{T_n})^p \int_{-1}^1 \text{up}(x)^p dx. \quad (3.9)$$

For every integer $m \geq 0$ and $n \geq 1$,

$$\int_{-1}^1 \text{up}^{(n)}(x)^m dx = \begin{cases} 0, & m \text{ odd}, \\ 2^{mT_n} \int_{-1}^1 \text{up}(x)^m dx, & m \text{ even}. \end{cases} \quad (3.10)$$

The absolute identity (3.9), with the exact hypothesis $p \geq 0$, is `intervalIntegral_abs_iteratedDeriv_rvachev_rpow`. At $p = 0$ this uses Lean's `Real.rpow` convention $0^0 = 1$. The signed parity simplification (3.10) is not separately formalized.

3.2 Every rearrangement-invariant norm scales exactly

A rearrangement-invariant quasi-norm on a finite measure space depends only on the distribution of $|f|$. This includes Lebesgue, Lorentz, Lorentz–Zygmund, Orlicz, and many interpolation spaces.

Theorem 3.3 (Universal norm ladder). *Let $X[-1, 1]$ be any absolutely homogeneous rearrangement-invariant quasi-normed function space. Then*

$$\left\| \text{up}^{(n)} \right\|_X = 2^{T_n} \left\| \text{up} \right\|_X, \quad n \geq 0. \quad (3.11)$$

In particular, for every $0 < p \leq \infty$,

$$\left\| \text{up}^{(n)} \right\|_{L^p(\mathbb{R})} = 2^{n(n+1)/2} \left\| \text{up} \right\|_{L^p(\mathbb{R})}. \quad (3.12)$$

Proof. The functions vanish outside $[-1, 1]$. By [Theorem 3.1](#), $2^{-T_n} |\text{up}^{(n)}|$ and up have the same decreasing rearrangement. Rearrangement invariance and absolute homogeneity give (3.11); the L^p case follows. $\square$

Function-space norm boundary. The real-power interval identity (3.9) is formal, but it is not yet packaged as a `Mathlib.eLpNorm` equality. The $p = \infty$ statement and the general rearrangement-invariant quasi-norm theorem (3.11) likewise have no matching Lean declarations. Consequently the universal norm ladder in this subsection remains a frontier layer; the Borel pushforward equality (3.8) is its completed measure-theoretic input.

Corollary 3.4 (Exact sup norms, total variations, and derivative ratios). *Since $\|\text{up}\|_1 = \|\text{up}\|_\infty = 1$,*

$$\left\| \text{up}^{(n)} \right\|_1 = 2^{T_n}, \quad (3.13)$$

$$\left\| \text{up}^{(n)} \right\|_\infty = 2^{T_n}, \quad (3.14)$$

$$\text{TV}(\text{up}^{(n)}) = \left\| \text{up}^{(n+1)} \right\|_1 = 2^{T_{n+1}}. \quad (3.15)$$

For every $0 < p \leq \infty$,

$$\frac{\left\| \text{up}^{(n+1)} \right\|_p}{\left\| \text{up}^{(n)} \right\|_p} = 2^{n+1}. \quad (3.16)$$

3.3 Thue–Morse extremizers and exact level sets

Corollary 3.5 (Midpoint extrema). *For $0 \leq k < 2^n$, put*

$$x_{n,k} := -1 + \frac{2k+1}{2^n}. \quad (3.17)$$

Then

$$\boxed{\text{up}^{(n)}(x_{n,k}) = 2^{T_n} \varepsilon_k.} \quad (3.18)$$

These are precisely the points at which $|\text{up}^{(n)}|$ attains its global maximum. Thus the signs of the complete extremal set of the n th derivative form the first 2^n terms of the signed Thue–Morse sequence.

Proof. Set $y = 0$ in (2.7) and use $\text{up}(0) = 1$. Strict decrease of up on $[0, 1]$ makes $y = 0$ the unique maximum inside each cell. $\square$

The signed midpoint equation (3.18) is formally `iteratedDeriv_rvachev_cell_zero`. This does not by itself formalize the exact inverse-Fabius level sets below.

Let $G = F^{-1} : [0, 1] \rightarrow [0, 1]$. Since $\text{up}(y) = 1 - F(|y|)$,

$$\{y \in [-1, 1] : \text{up}(y) > t\} = (-G(1-t), G(1-t)), \quad 0 < t < 1. \quad (3.19)$$

Theorem 3.6 (Exact inverse-Fabius level-set law). *For every $n \geq 0$ and $0 < t < 1$,*

$$\boxed{\left| \left\{ x \in \mathbb{R} : \left| \text{up}^{(n)}(x) \right| > 2^{T_n} t \right\} \right| = 2G(1-t).} \quad (3.20)$$

Consequently, if $V_n = 2^{-T_n} |\text{up}^{(n)}(U)|$, then

$$\mathbb{P}(V_n > t) = G(1-t), \quad (3.21)$$

$$\mathbb{P}(V_n \leq t) = 1 - G(1-t), \quad (3.22)$$

$$f_{V_n}(t) = G'(1-t), \quad 0 < t < 1. \quad (3.23)$$

For $n \geq 1$, the signed variable $2^{-T_n} \text{up}^{(n)}(U)$ has density

$$f_n(v) = \frac{1}{2} G'(1-|v|), \quad -1 < v < 1. \quad (3.24)$$

Proof. Combine equimeasurability with (3.19). The density follows by differentiating the distribution function on the open interval. The signed density is the symmetric Rademacher mixture. $\square$

Closed-form level-set boundary. Equation (3.8) proves exact absolute equimeasurability as a Borel measure identity. The further identifications (3.20)–(3.24), which express the sets and densities explicitly through the totalized inverse G , do not yet have matching Lean theorems. In particular, the exact level-set and density package remains unformalized.

Remark 3.7 (A quantile coupling). If U is uniform on $[-1, 1]$, then $|U|$ is uniform on $[0, 1]$ and

$$G(1 - \text{up}(U)) = |U|. \quad (3.25)$$

Thus the inverse Fabius function is not merely asymptotically related to derivative values; it is their exact probability integral transform.

4 The norm spectrum determines the Fabius function

Define the normalized norm spectrum

$$A_p := \frac{1}{2} \|\text{up}\|_p^p = 2^{-1-pT_n} \left\| \text{up}^{(n)} \right\|_p^p, \quad p > 0. \quad (4.1)$$

The second expression is independent of n by (3.12). Set $A_0 := 1$. If $W = \text{up}(U)$, then $A_p = \mathbb{E}W^p$ for every $p \geq 0$.

Proposition 4.1 (Complete monotonicity and log-convexity). *The function $p \mapsto A_p$ is strictly completely monotone on $(0, \infty)$:*

$$(-1)^r A_p^{(r)} = \mathbb{E}[W^p (-\log W)^r] > 0, \quad r \geq 0. \quad (4.2)$$

It is strictly log-convex. Its integer samples are a completely monotone Hausdorff moment sequence:

$$(-1)^r \Delta^r A_m = \mathbb{E}[W^m (1 - W)^r] > 0, \quad m, r \geq 0, \quad (4.3)$$

where $\Delta A_m = A_{m+1} - A_m$.

Proof. Write $W^p = \exp(-pX)$ with $X = -\log W \geq 0$ and differentiate under the integral. Strict log-convexity follows from Hölder's inequality because W is nonconstant. The finite-difference formula follows by expanding $(1 - W)^r$. $\square$

Theorem 4.2 (Hausdorff reconstruction from integer derivative norms). *The sequence $\{A_m\}_{m \geq 0}$ uniquely determines G , hence F and up . More explicitly, define*

$$b_{N,k} := \binom{N}{k} \sum_{j=0}^{N-k} (-1)^j \binom{N-k}{j} A_{k+j}, \quad 0 \leq k \leq N. \quad (4.4)$$

Then $b_{N,k} \geq 0$, $\sum_k b_{N,k} = 1$, and at every $0 < t < 1$,

$$G(1 - t) = 1 - \lim_{N \rightarrow \infty} \sum_{k=0}^{\lfloor Nt \rfloor} b_{N,k}. \quad (4.5)$$

The same reconstruction applies to the normalized norms of any fixed derivative $\text{up}^{(n)}$ through (4.1).

Proof. By (4.3),

$$b_{N,k} = \binom{N}{k} \mathbb{E}[W^k (1 - W)^{N-k}].$$

Conditioned on $W = w$, these are the probabilities of a binomial random variable $K_N \sim \text{Bin}(N, w)$. Therefore

$$\sum_{k=0}^{\lfloor Nt \rfloor} b_{N,k} = \mathbb{E}\mathbb{P}(K_N/N \leq t \mid W).$$

The conditional law of large numbers makes the inner probability converge to $\mathbf{1}_{\{W \leq t\}}$ away from $w = t$. The distribution of W is continuous, so the expectation converges to $\mathbb{P}(W \leq t) = 1 - G(1 - t)$ by (3.22). Uniqueness is also the classical determinacy of the Hausdorff moment problem on $[0, 1]$. $\square$

Remark 4.3. This theorem gives a perhaps unexpected completeness statement: the scalar sequence $\|\text{up}^{(n)}\|_m$, $m = 1, 2, \dots$, for any one derivative order n , contains the entire Fabius function. The Thue–Morse oscillation disappears under absolute moments, yet no information about the underlying profile is lost.

5 An exact inverse-Fabius beta transform

Theorem 5.1 (Beta transform of the inverse Fabius function). *For every real $p > 0$,*

$$A_p = p \int_0^1 (1-y)^{p-1} G(y) \, dy. \quad (5.1)$$

Equivalently, if $Y_p \sim \text{Beta}(1, p)$, then

$$A_p = \mathbb{E}G(Y_p). \quad (5.2)$$

Proof. Using evenness and (2.2),

$$A_p = \int_0^1 (1-F(x))^p \, dx.$$

The monotone substitution $y = F(x)$, interpreted as a Lebesgue–Stieltjes substitution at the flat endpoints, gives

$$A_p = \int_0^1 (1-y)^p \, dG(y).$$

Integration by parts has no boundary term because $G(0) = 0$ and $(1-y)^p = 0$ at $y = 1$. This proves (5.1); the beta density is $p(1-y)^{p-1}$. $\square$

Corollary 5.2 (Layer-cake form). *For every $p > 0$ and every $n \geq 0$,*

$$\left\| \text{up}^{(n)} \right\|_p^p = 2^{pT_n+1} p \int_0^1 (1-y)^{p-1} G(y) \, dy. \quad (5.3)$$

The beta kernel concentrates at $y \asymp p^{-1}$. Because G is slowly varying at zero, (5.1) already implies $A_p \sim G(1/p)$. The next section replaces that first-order Abelian statement by an all-orders differential operator.

6 The Euler–Gamma differential operator

6.1 Exact Mellin symbol

Let

$$\mathcal{D} := y \frac{d}{dy} \quad (6.1)$$

be the Euler dilation derivative. Put $Z_p = pY_p$.

Proposition 6.1 (Mellin symbol of the beta dilation). *For $\Re s > -1$,*

$$M_p(s) := \mathbb{E}Z_p^s = p^s \frac{\Gamma(p+1)\Gamma(1+s)}{\Gamma(p+1+s)}. \quad (6.2)$$

For any bounded extension of G to $(1, \infty)$,

$$A_p = [M_p(\mathcal{D})G](p^{-1}), \quad (6.3)$$

where the operator is rigorously interpreted by the Bochner average

$$[M_p(\mathcal{D})H](y) := \mathbb{E}[H(Z_p y)]. \quad (6.4)$$

The value in (6.3) is independent of how G is extended, because $0 < Z_p/p = Y_p < 1$.

Proof. The beta integral gives

$$\mathbb{E}Z_p^s = p^s p \int_0^1 y^s (1-y)^{p-1} dy = p^s \frac{\Gamma(p+1)\Gamma(1+s)}{\Gamma(p+1+s)}.$$

The dilation semigroup satisfies $\exp(t\mathcal{D})H(y) = H(e^t y)$. Averaging with $t = \log Z_p$ and evaluating at $y = p^{-1}$ gives $\mathbb{E}G(Z_p/p) = \mathbb{E}G(Y_p) = A_p$. $\square$

As $p \rightarrow \infty$, Z_p converges to an exponential random variable Z of mean one, and $M_p(s) \rightarrow \Gamma(1+s)$. Thus $\Gamma(1+\mathcal{D})$ is not decorative notation: it is the Mellin symbol of the limiting random dilation.

6.2 Cumulants, Bell polynomials, and finite- p corrections

Let

$$K_p(s) := \log M_p(s). \quad (6.5)$$

The cumulants of $\log Z_p$ are

$$\kappa_{1,p} = \log p - \psi(p+1) - \gamma, \quad (6.6)$$

$$\kappa_{m,p} = \psi^{(m-1)}(1) - \psi^{(m-1)}(p+1), \quad m \geq 2. \quad (6.7)$$

If B_m denotes the complete exponential Bell polynomial, then

$$\mu_{m,p} := \mathbb{E}(\log Z_p)^m = B_m(\kappa_{1,p}, \dots, \kappa_{m,p}). \quad (6.8)$$

As $p \rightarrow \infty$,

$$\kappa_1 = -\gamma, \quad (6.9)$$

$$\kappa_m = (-1)^m (m-1)! \zeta(m), \quad m \geq 2, \quad (6.10)$$

which are the cumulants of $\log Z$ for $Z \sim \text{Exp}(1)$.

The uniform gamma-ratio expansion on compact s -sets is

$$K_p(s) \sim \log \Gamma(1+s) + \sum_{r=1}^{\infty} \frac{(-1)^{r+1} (B_{r+1}(1) - B_{r+1}(1+s))}{r(r+1)p^r}. \quad (6.11)$$

Here $B_m(x)$ are Bernoulli polynomials. The first correction is $-s(s+1)/(2p)$.

Theorem 6.2 (Local Gamma–Bell expansion). *Fix $N \geq 1$. As $p \rightarrow \infty$,*

$$A_p = \sum_{m=0}^{N-1} \frac{\mu_m}{m!} \mathcal{D}^m G(p^{-1}) + \mathcal{O}_N \left(G(p^{-1}) \frac{(\log \rho)^{c_N}}{\rho^N} \right), \quad (6.12)$$

where

$$\rho = \sqrt{\frac{2 \log p}{\log 2}}, \quad \mu_m = B_m(-\gamma, \zeta(2), -2\zeta(3), 6\zeta(4), \dots), \quad (6.13)$$

and c_N is a finite integer depending only on N . A finite- p expansion with corrections to every power of p^{-1} is obtained by replacing μ_m with (6.8) or, equivalently, substituting (6.11) into $\exp K_p(\mathcal{D})$.

Proof. Write $y = p^{-1}$ and $t = \log Z_p$. On a logarithmic neighborhood of y ,

$$G(e^t y) = \sum_{m=0}^{N-1} \frac{t^m}{m!} \mathcal{D}^m G(y) + R_N(t, y).$$

The all-orders inverse endpoint expansion in the corpus implies, for each fixed m , $\mathcal{D}^m G(y)/G(y)$ is bounded by a power of $\log \rho$ times ρ^{-m} . Fix a constant $C > L + 1$ and split the expectation into $|t| \leq C\rho$ and its complement. On the central range, every point $e^u y$ between y and $e^t y$ has endpoint coordinate $\rho + \mathcal{O}_C(1)$. Integrating the elasticity shows $G(e^u y)/G(y) = \mathcal{O}_C(1)$ there, so Taylor's integral remainder is

$$|R_N(t, y)| \leq C_N G(y) \frac{(\log \rho)^{c_N}}{\rho^N} |t|^N.$$

The moments $\mathbb{E}|\log Z_p|^N$ are bounded uniformly for $p \geq 2$: near zero the density of Z_p is at most one, while for $z \geq 1$ it is bounded by $e^{-z/2}$. For the omitted negative tail, $\mathbb{P}(Z_p < e^{-C\rho}) \leq e^{-C\rho}$; for the positive tail, $\mathbb{P}(Z_p > e^{C\rho}) \leq \exp(-e^{C\rho})$ once p is large. Since $G(p^{-1}) \asymp \rho e^{-L\rho}$, both tail contributions are $o(G(p^{-1})\rho^{-N})$. Taking expectations gives coefficients $\mu_{m,p}$; replacing them by μ_m changes the result by $\mathcal{O}(p^{-1}G(p^{-1}))$, which is beyond every power of ρ^{-1} relative to $G(p^{-1})$. The Bell and Bernoulli descriptions follow from differentiation of (6.5) and the standard gamma-ratio expansion. $\square$

The first five log-exponential moments are

$$\begin{aligned} \mu_0 &= 1, \\ \mu_1 &= -\gamma, \\ \mu_2 &= \gamma^2 + \zeta(2), \\ \mu_3 &= -\gamma^3 - 3\gamma\zeta(2) - 2\zeta(3), \\ \mu_4 &= \gamma^4 + 6\gamma^2\zeta(2) + 8\gamma\zeta(3) + 3\zeta(2)^2 + 6\zeta(4). \end{aligned} \tag{6.14}$$

Thus

$$\begin{aligned} A_p &= G - \gamma \mathcal{D}G + \frac{\gamma^2 + \pi^2/6}{2} \mathcal{D}^2 G \\ &\quad - \frac{\gamma^3 + \gamma\pi^2/2 + 2\zeta(3)}{6} \mathcal{D}^3 G \\ &\quad + \frac{\gamma^4 + \gamma^2\pi^2 + 8\gamma\zeta(3) + 3\pi^4/20}{24} \mathcal{D}^4 G + \dots, \quad y = p^{-1}. \end{aligned} \tag{6.15}$$

Remark 6.3 (Why Bell and Bernoulli polynomials appear). Bell polynomials convert the cumulants of the random logarithmic dilation into its ordinary moments, hence into powers of $\mathcal{D}$. Bernoulli polynomials govern the finite- p deviation of the beta dilation from the limiting exponential dilation through the quotient $\Gamma(p+1)/\Gamma(p+1+s)$. The two families enter for different structural reasons and then compose automatically.

7 Large-p norms, Lambert coordinates, and phase tomography

7.1 Imported inverse endpoint expansion

Put

$$L := \log 2, \quad T := \log p, \quad \rho := \sqrt{\frac{2T}{L}}, \quad \beta := \frac{1}{L} + \frac{1}{2}, \quad \theta := \rho + \beta. \tag{7.1}$$

The inverse-frontier report in the corpus proves

$$G_0(p^{-1}) := \rho e^{-L\rho - L\beta} = \frac{\sqrt{T}}{e\sqrt{L}} e^{-\sqrt{2LT}}, \tag{7.2}$$

and

$$\frac{G(p^{-1})}{G_0(p^{-1})} = 1 + \frac{\frac{1}{2} \log \rho + \kappa_0 - \Psi(\theta)}{\rho} + \mathcal{O}\left(\frac{(\log \rho)^2}{\rho^2}\right), \tag{7.3}$$

where Ψ is a nonconstant real-analytic one-periodic Gamma-zeta function. It also proves the endpoint elasticity

$$\frac{yG'(y)}{G(y)} = \frac{1}{\rho} + \frac{\Psi'(\rho + \beta) - 1}{L\rho^2} + \mathcal{O}\left(\frac{\log \rho}{\rho^3}\right), \quad y \downarrow 0. \quad (7.4)$$

These are baseline inputs; the norm consequences below are new.

7.2 Euler's constant shifts the inverse correction

Theorem 7.1 (Two-term norm-to-quantile ratio). *As $p \rightarrow \infty$,*

$$\boxed{\frac{A_p}{G(p^{-1})} = 1 - \frac{\gamma}{\rho} + \frac{1}{\rho^2} \left[\frac{\gamma^2 + \pi^2/6}{2} - \frac{\gamma(\Psi'(\theta) - 1)}{L} \right] + \mathcal{O}\left(\frac{\log \rho}{\rho^3}\right)}. \quad (7.5)$$

In particular,

$$\boxed{A_p \sim G(p^{-1})}. \quad (7.6)$$

Proof. By (6.15),

$$\frac{A_p}{G} = 1 - \gamma \frac{\mathcal{D}G}{G} + \frac{\gamma^2 + \pi^2/6}{2} \frac{\mathcal{D}^2 G}{G} + \mathcal{O}(\rho^{-3} \log \rho).$$

Equation (7.4) gives

$$\frac{\mathcal{D}G}{G} = \frac{1}{\rho} + \frac{\Psi'(\theta) - 1}{L\rho^2} + \mathcal{O}\left(\frac{\log \rho}{\rho^3}\right).$$

If $a = \mathcal{D} \log G$, then $\mathcal{D}^2 G/G = a^2 + \mathcal{D}a = \rho^{-2} + \mathcal{O}((\log \rho)/\rho^3)$. Substitution proves the formula. $\square$

Corollary 7.2 (First periodic norm expansion and Euler shift). *As $p \rightarrow \infty$,*

$$\boxed{\frac{\|\text{up}\|_p^p}{2G_0(p^{-1})} = 1 + \frac{\frac{1}{2} \log \rho + \kappa_0 - \Psi(\theta) - \gamma}{\rho} + \mathcal{O}\left(\frac{(\log \rho)^2}{\rho^2}\right)}. \quad (7.7)$$

Thus the first correction for the global L^p norm is the first inverse-quantile correction with the universal subtraction γ .

Proof. Multiply (7.5) by (7.3) and use $\|\text{up}\|_p^p = 2A_p$. $\square$

Corollary 7.3 (Approach to the sup norm). *As $p \rightarrow \infty$,*

$$\boxed{1 - \|\text{up}\|_p \sim \frac{\sqrt{2 \log 2 \log p}}{p}}. \quad (7.8)$$

More precisely,

$$-p \log \|\text{up}\|_p = L\rho - \log \rho + 1 - \frac{L}{2} - \frac{\frac{1}{2} \log \rho + \kappa_0 - \Psi(\theta) - \gamma}{\rho} + \mathcal{O}\left(\frac{(\log \rho)^2}{\rho^2}\right). \quad (7.9)$$

Proof. Take logarithms in (7.7) and use $G_0 = \rho \exp(-L\rho - L\beta)$ with $L\beta = 1 + L/2$. The leading term of $-\log \|\text{up}\|_p$ is $L\rho/p$; since it tends to zero, $1 - \|\text{up}\|_p \sim -\log \|\text{up}\|_p$. $\square$

7.3 Recovering the endpoint periodic function from global norms

Theorem 7.4 (Phase tomography by L^p norms). *Fix a phase $\vartheta \in \mathbb{R}/\mathbb{Z}$. For all sufficiently large integers j , define*

$$\rho_j := j + \vartheta - \beta, \quad p_j := \exp\left(\frac{L}{2}\rho_j^2\right). \quad (7.10)$$

Then

$$\lim_{j \rightarrow \infty} \left[\rho_j \left(\frac{\|\text{up}\|_{p_j}^{p_j}}{2G_0(p_j^{-1})} - 1 \right) - \frac{1}{2} \log \rho_j \right] = \kappa_0 - \gamma - \Psi(\vartheta). \quad (7.11)$$

Hence the nonconstant endpoint fluctuation Ψ can be reconstructed from a family of global norm measurements; pointwise evaluation or inversion of F is not required.

Proof. For (7.10), the parameter in (7.1) is exactly ρ_j and $\theta_j = \rho_j + \beta = j + \vartheta$. Periodicity gives $\Psi(\theta_j) = \Psi(\vartheta)$. Multiply (7.7) by ρ_j and subtract $\frac{1}{2} \log \rho_j$. $\square$

Remark 7.5 (Lambert interpretation). The Gaussian-logarithmic sampling law $\log p_j = (L/2)\rho_j^2$ is the inverse-side analogue of the lower-Lambert phase coordinate used in the endpoint expansion. It is precisely the sampling geometry that freezes the periodic saddle phase while sending the asymptotic scale to zero.

8 Fourier energy and a shifted Stieltjes–Wigert moment problem

8.1 Exact spectral moments

Let

$$C_2 := \|\text{up}\|_2^2 = \int_{\mathbb{R}} |\Phi(\xi)|^2 d\xi. \quad (8.1)$$

Numerically, $C_2 = 0.8088735359679707\dots$, but no decimal evaluation is needed below.

Theorem 8.1 (Shifted Stieltjes–Wigert spectral moments). *Define the probability measure*

$$d\nu(\xi) := \frac{|\Phi(\xi)|^2}{C_2} d\xi \quad (8.2)$$

and let μ_{sp} be its pushforward by $z = (2\pi\xi)^2$. Then, for every $n \geq 0$,

$$\int_0^\infty z^n d\mu_{\text{sp}}(z) = 2^{n(n+1)}. \quad (8.3)$$

With $q = 1/4$, this is

$$m_n = q^{-n(n+1)/2}. \quad (8.4)$$

Proof. Parseval and the Fourier differentiation rule give

$$\int_{\mathbb{R}} (2\pi\xi)^{2n} |\Phi(\xi)|^2 d\xi = \left\| \text{up}^{(n)} \right\|_2^2.$$

By (3.12) with $p = 2$, the right side is $2^{2T_n} C_2 = 2^{n(n+1)} C_2$. Divide by C_2 and push forward. $\square$

The measure has the explicit density

$$w_{\text{sp}}(z) = \frac{|\Phi(\sqrt{z}/(2\pi))|^2}{2\pi C_2 \sqrt{z}}, \quad z > 0. \quad (8.5)$$

8.2 A direct proof of moment indeterminacy

Let $L = \log 2$ and define the lognormal density

$$w_{\ln}(z) := \frac{1}{z\sqrt{4\pi L}} \exp\left(-\frac{(\log z - L)^2}{4L}\right), \quad z > 0. \quad (8.6)$$

This is the law of $\exp N(L, 2L)$ and therefore

$$\int_0^\infty z^n w_{\ln}(z) dz = \exp(nL + n^2 L) = 2^{n(n+1)}. \quad (8.7)$$

Corollary 8.2 (Explicit indeterminacy). *The Stieltjes moment sequence (8.3) is indeterminate. Indeed, $w_{\text{sp}}(z) dz$ and $w_{\ln}(z) dz$ are distinct probability measures with all moments equal. More generally,*

$$w_t := (1-t)w_{\ln} + tw_{\text{sp}}, \quad 0 \leq t \leq 1, \quad (8.8)$$

forms an explicit continuum of representing densities.

Proof. The moment equality is (8.3)–(8.7). The densities are different because w_{sp} vanishes whenever $z = (2\pi m)^2$, $m \in \mathbb{N}$, whereas $w_{\ln}$ is strictly positive on $(0, \infty)$. Convex combinations preserve every moment. $\square$

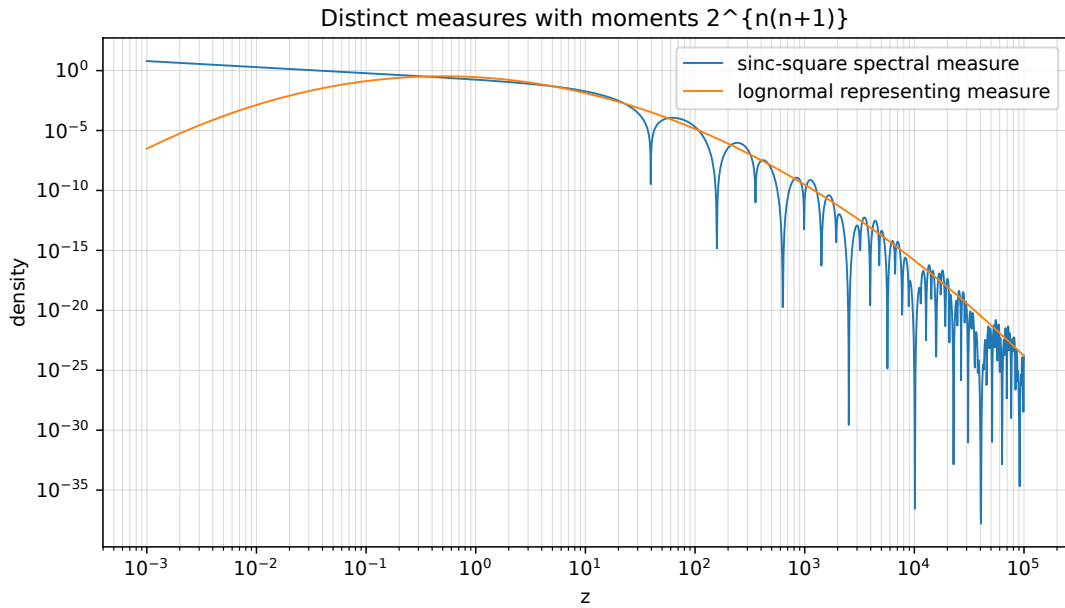


Figure 1: The sinc-square spectral density and a lognormal density have exactly the same nonnegative integer moments but visibly different local structures. The spectral zeros are inherited from the integer zeros of the infinite sinc product.

8.3 A nonclassical q-Pearson equation

The Fourier product satisfies

$$\Phi(2\xi) = \text{sinc}(2\pi\xi)\Phi(\xi). \quad (8.9)$$

Proposition 8.3 (Sinc-square q-Pearson law). *The spectral density (8.5) obeys*

$$w_{\text{sp}}(4z) = \frac{1}{2} \text{sinc}^2(\sqrt{z}) w_{\text{sp}}(z). \quad (8.10)$$

By contrast, the lognormal representative satisfies

$$w_{\ln}(4z) = \frac{1}{4z} w_{\ln}(z). \quad (8.11)$$

Proof. In (8.5), replacing z by $4z$ doubles the Fourier argument and multiplies the square-root denominator by 2. Apply (8.9) and note that $2\pi(\sqrt{z}/(2\pi)) = \sqrt{z}$. The lognormal identity follows by direct substitution in (8.6). $\square$

Remark 8.4. Classical Stieltjes–Wigert measures satisfy q-Pearson equations with monomial or rational multipliers. Equation (8.10) replaces that multiplier by an entire sinc-square factor carrying the complete integer-zero divisor. Determining its position in the Nevanlinna family is one of the principal open directions in Section 12.

9 Hankel determinants, q-binomial polynomials, and Lagrange quadrature

Fix $0 < q < 1$ and the moment sequence

$$m_n = q^{-n(n+1)/2}. \quad (9.1)$$

The Fabius–Rvachev spectral case is $q = 1/4$.

Theorem 9.1 (Hankel determinants). *For $N \geq 0$,*

$$\Delta_N := \det[m_{i+j}]_{i,j=0}^N = q^{-N(N+1)(4N+5)/6} \prod_{k=1}^N (q; q)_k, \quad (9.2)$$

where $(q; q)_k = \prod_{r=1}^k (1 - q^r)$.

Proof. Factor

$$m_{i+j} = q^{-i(i+1)/2} q^{-j(j+1)/2} q^{-ij}.$$

The row and column factors contribute $q^{-\sum_{i=0}^N i(i+1)}$. The remaining determinant is the Vandermonde determinant in q^{-i} :

$$\det[(q^{-i})^j]_{i,j=0}^N = \prod_{0 \leq i < k \leq N} (q^{-k} - q^{-i}) = q^{-\sum_{k=1}^N k^2} \prod_{k=1}^N (q; q)_k.$$

Finally

$$\sum_{i=1}^N i(i+1) + \sum_{i=1}^N i^2 = \frac{N(N+1)(4N+5)}{6}.$$

$\square$

Theorem 9.2 (Monic shifted Stieltjes–Wigert polynomials). *The monic orthogonal polynomial of degree n for any representing measure of (9.1) is*

$$P_n(z) = \sum_{k=0}^n (-1)^{n-k} q^{k^2 - n^2} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k. \quad (9.3)$$

It satisfies

$$\int z^j P_n(z) d\mu(z) = 0, \quad 0 \leq j < n, \quad (9.4)$$

for every measure μ with moments (9.1), and

$$h_n := \int P_n(z)^2 d\mu(z) = q^{-(2n^2+n)}(q; q)_n. \quad (9.5)$$

The three-term recurrence is

$$P_{n+1}(z) = (z - a_n)P_n(z) - b_nP_{n-1}(z), \quad (9.6)$$

with

$$a_n = q^{-2n-1}(1 + q - q^{n+1}), \quad (9.7)$$

$$b_n = q^{-(4n-1)}(1 - q^n), \quad n \geq 1. \quad (9.8)$$

Proof. Let c_k be the coefficient of z^k in (9.3). For $j < n$,

$$\begin{aligned} \sum_{k=0}^n c_k m_{k+j} &= (-1)^n q^{-n^2-j(j+1)/2} \sum_{k=0}^n (-1)^k q^{k(k-1)/2} \begin{bmatrix} n \\ k \end{bmatrix}_q q^{-jk} \\ &= (-1)^n q^{-n^2-j(j+1)/2} (q^{-j}; q)_n = 0, \end{aligned}$$

by the finite q-binomial theorem; one factor is $1 - q^0$. The leading coefficient is one. The norm is the standard Gram-determinant quotient $h_n = \Delta_n/\Delta_{n-1}$, which gives (9.5). Hence $b_n = h_n/h_{n-1}$. Comparing the coefficients of z^n in the monic recurrence gives a_n ; the coefficient of z^{n-1} in P_n is $-q^{-2n+1}(1 - q^n)/(1 - q)$, yielding (9.7). $\square$

For $q = 1/4$, the first polynomials are

$$\begin{aligned} P_0(z) &= 1, \\ P_1(z) &= z - 4, \\ P_2(z) &= z^2 - 80z + 256, \\ P_3(z) &= (z - 64)(z^2 - 1280z + 4096). \end{aligned} \quad (9.9)$$

9.1 Gaussian quadrature as q-binomial Lagrange interpolation

Let $\lambda_{n,1}, \dots, \lambda_{n,n}$ be the zeros of P_n and let $\ell_{n,j}$ be the Lagrange cardinal polynomial at these nodes. The Gaussian weights are

$$\omega_{n,j} := \int \ell_{n,j}(z) d\mu(z) = \left(\sum_{r=0}^{n-1} \frac{P_r(\lambda_{n,j})^2}{h_r} \right)^{-1}. \quad (9.10)$$

Because the polynomials and h_r are determined by the moment sequence, the nodes and weights are identical for μ_{sp} , the lognormal measure, and every other representing measure.

Corollary 9.3 (Measure-independent Gaussian quadrature). *For every polynomial Q of degree at most $2n - 1$,*

$$\int_0^\infty Q(z) d\mu_{\text{sp}}(z) = \sum_{j=1}^n \omega_{n,j} Q(\lambda_{n,j}) = \int_0^\infty Q(z) w_{\text{ln}}(z) dz. \quad (9.11)$$

The q-binomial coefficients in (9.3) therefore generate exact finite Lagrange quadrature rules for polynomial spectral energies of the infinite sinc product.

Remark 9.4. Moment indeterminacy is invisible to every finite polynomial test: all representing measures produce the same Gaussian quadrature. The distinction lives in the orthogonal complement of the polynomial closure, exactly where the integer-zero geometry of Φ can enter.

10 The optimal derivative weight and a q-Legendre Fourier envelope

10.1 Exact Denjoy–Carleman weight

Define

$$M_n := 2^{T_n} = 2^{n(n+1)/2}. \quad (10.1)$$

By (3.14), up saturates this derivative weight exactly:

$$\left\| \text{up}^{(n)} \right\|_{\infty} = M_n. \quad (10.2)$$

The sequence is logarithmically convex and

$$\frac{M_n}{M_{n+1}} = 2^{-(n+1)}, \quad \sum_{n=0}^{\infty} \frac{M_n}{M_{n+1}} = 1 < \infty. \quad (10.3)$$

Thus the direct-weight Denjoy–Carleman class generated by M is nonquasianalytic, as the existence of the compactly supported nonzero function up requires.

Proposition 10.1 (Outside every finite Gevrey class). *For no finite $s > 0$ do constants $A, C > 0$ satisfy*

$$M_n \leq CA^n (n!)^s \quad (n \geq 0). \quad (10.4)$$

Hence up belongs to no finite-order Gevrey class on any interval containing its support.

Proof. The logarithm of M_n is $(\log 2)n^2/2 + \mathcal{O}(n)$, whereas $\log((n!)^s A^n) = sn \log n + \mathcal{O}(n)$. The quadratic term wins. $\square$

10.2 Exact associated function and periodic lattice correction

Define the associated function

$$\omega_M(t) := \sup_{n \geq 0} \log \frac{t^n}{M_n}, \quad t > 0. \quad (10.5)$$

Let $r = \log_2 t$.

Theorem 10.2 (Exact q-Legendre transform). *For $r \geq 1/2$,*

$$\omega_M(t) = \frac{\log 2}{2} \left[\left(r - \frac{1}{2} \right)^2 - \text{dist} \left(r - \frac{1}{2}, \mathbb{N}_0 \right)^2 \right]. \quad (10.6)$$

For $r \leq 1/2$, $\omega_M(t) = 0$. Thus the continuous log-square Legendre transform carries an exact bounded one-periodic lattice correction in $\log_2 t$.

Proof. For each integer $n \geq 0$,

$$\log \frac{t^n}{M_n} = \log 2 \left(nr - \frac{n(n+1)}{2} \right) = \frac{\log 2}{2} \left[\left(r - \frac{1}{2} \right)^2 - \left(n - r + \frac{1}{2} \right)^2 \right].$$

Maximizing means choosing the nearest nonnegative integer to $r - 1/2$. If $r \leq 1/2$, the maximum is attained at $n = 0$ and equals zero. $\square$

Corollary 10.3 (Optimized integration-by-parts envelope). *For every $\xi \neq 0$,*

$$|\Phi(\xi)| \leq \inf_{n \geq 0} \frac{M_n}{(2\pi|\xi|)^n} = \exp(-\omega_M(2\pi|\xi|)). \quad (10.7)$$

If $r = \log_2(2\pi|\xi|) \geq 1/2$, this is

$$|\Phi(\xi)| \leq 2^{-\frac{1}{2}(r-1/2)^2 + \frac{1}{2} \text{dist}(r-1/2, \mathbb{N}_0)^2}. \quad (10.8)$$

Proof. Integrating the Fourier transform by parts n times and using (3.13) gives $|\Phi(\xi)| \leq M_n/(2\pi|\xi|)^n$. Optimize with Theorem 10.2. $\square$

Warning 10.4. The envelope (10.8) is the best one obtainable from the exact L^1 derivative norms alone; it is not the sharp pointwise or annular Fourier envelope. The Fourier-decay reports in the corpus exploit the lacunary sine cocycle and obtain stronger linear-in-log $|\xi|$ corrections. The value of (10.8) is its exact functional-analytic duality with the derivative weight, not a replacement for those sharper results.

11 Numerical experiments

All code used here is in `numerical_experiments.py`. It is included with detailed comments in the accompanying archive. The script reconstructs `up` by inverse FFT on a periodic box of length 4 from 55 dyadic sinc factors, using 2^{19} samples. Since `up` is supported on $[-1, 1]$, the box supplies exact geometric zero padding. The run used

$$\Delta x = 2^{-17} = 7.62939453125 \times 10^{-6}, \quad \Delta \xi = \frac{1}{4}.$$

The reconstructed density had integral 1 and value $\text{up}(0) = 1$ to displayed precision; the largest value outside the support was 3.33×10^{-16} .

11.1 Derivative tiling and norm checks

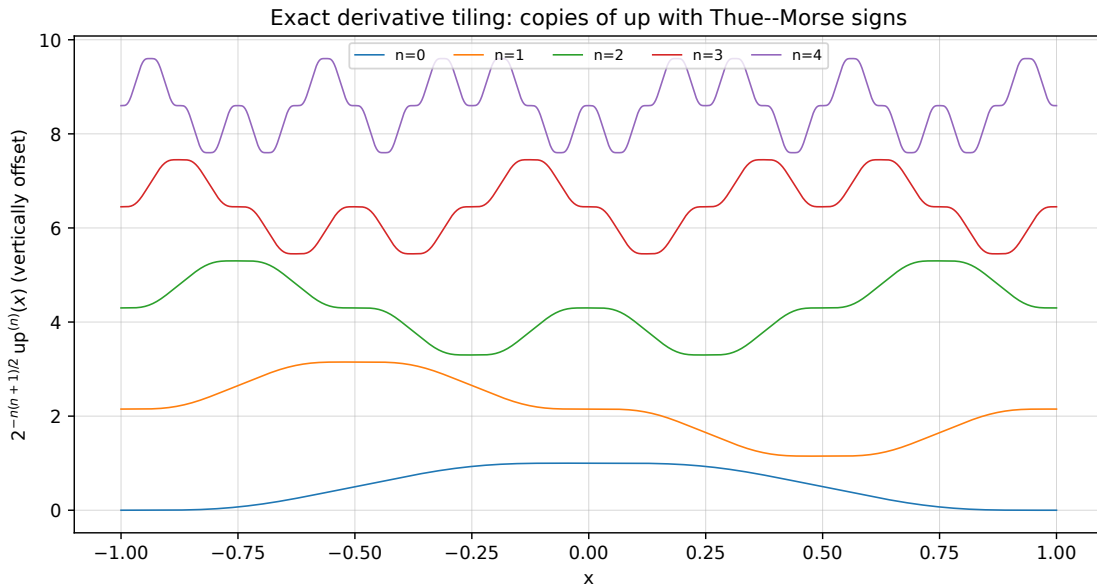


Figure 2: Normalized derivatives $2^{-T_n} \text{up}^{(n)}$, vertically offset. Each row is a concatenation of copies of the original bump; their signs are the length- 2^n Thue-Morse block.

For $n = 0, \dots, 6$ and $p \in \{1, 2, 4, 8, \infty\}$, the script formed the derivative from the exact cell formula and compared its numerical norm with $2^{T_n} \|\text{up}\|_p$. The largest absolute deviation of the ratio from 1 was 2.22×10^{-16} .

11.2 Spectral moments

n	numerical $\int (2\pi\xi)^{2n} \Phi ^2$	predicted $2^{n(n+1)} C_2$	relative error
0	$8.088735359679706 \times 10^{-1}$	$8.088735359679707 \times 10^{-1}$	-1.11×10^{-16}
1	3.235494143871882	3.235494143871883	-2.22×10^{-16}
2	$5.176790630195011 \times 10^1$	$5.176790630195013 \times 10^1$	-2.22×10^{-16}
3	$3.313146003324810 \times 10^3$	$3.313146003324808 \times 10^3$	4.44×10^{-16}
4	$8.481653768511517 \times 10^5$	$8.481653768511509 \times 10^5$	8.88×10^{-16}
5	$8.685213458955796 \times 10^8$	$8.685213458955785 \times 10^8$	1.33×10^{-15}
6	$3.557463432788293 \times 10^{12}$	$3.557463432788290 \times 10^{12}$	8.88×10^{-16}
7	$5.828548088280333 \times 10^{16}$	$5.828548088280334 \times 10^{16}$	-1.11×10^{-16}

The near-machine equality is a stringent joint check of the FFT normalization, derivative scale, Parseval convention, and moment exponent.

11.3 Euler–Gamma operator convergence

The inverse interpolant was obtained directly from $G(1 - \text{up}(x)) = x$ on $0 \leq x \leq 1$. Logarithmic finite differences estimated $\mathcal{D}^k G(1/p)$ for $k \leq 3$. The following table lists relative errors of successive truncations of (6.15).

p	$A_p/G(1/p)$	G only	through $\mathcal{D}$	through $\mathcal{D}^2$	through $\mathcal{D}^3$
256	0.92618125	-7.38×10^{-2}	4.84×10^{-2}	-1.47×10^{-2}	6.46×10^{-3}
512	0.92793214	-7.21×10^{-2}	4.32×10^{-2}	-1.20×10^{-2}	6.37×10^{-3}
1024	0.92942377	-7.06×10^{-2}	3.91×10^{-2}	-9.78×10^{-3}	4.87×10^{-3}
2048	0.93089393	-6.91×10^{-2}	3.60×10^{-2}	-8.37×10^{-3}	3.35×10^{-3}
4096	0.93235311	-6.76×10^{-2}	3.33×10^{-2}	-7.55×10^{-3}	2.53×10^{-3}
8192	0.93370939	-6.63×10^{-2}	3.09×10^{-2}	-6.86×10^{-3}	2.39×10^{-3}
16384	0.93492704	-6.51×10^{-2}	2.87×10^{-2}	-6.10×10^{-3}	2.29×10^{-3}
32768	0.93604971	-6.40×10^{-2}	2.69×10^{-2}	-5.39×10^{-3}	1.97×10^{-3}
65536	0.93712995	-6.29×10^{-2}	2.53×10^{-2}	-4.83×10^{-3}	1.60×10^{-3}

11.4 Symbolic Hankel checks

SymPy evaluated the determinants in (9.2) exactly for $N = 0, \dots, 6$ at $q = 1/4$; every quotient of the direct determinant by the claimed closed form simplified to 1. The base-10 logarithms of the determinants for $N = 1, \dots, 6$ were

$$1.68124, 7.54887, 20.03233, 41.54498, 74.49635, 121.29499,$$

illustrating the rapid $q^{-\frac{2}{3}N^3 + \mathcal{O}(N^2)}$ growth.

12 Conjectures and promising research directions

12.1 A full norm transseries

The inverse report gives an all-orders expansion of G in the differential algebra generated by periodic saddle jets. The exact operator (6.3) should transport that algebra to the norm spectrum without introducing any new transcendental data beyond the log-exponential cumulants.

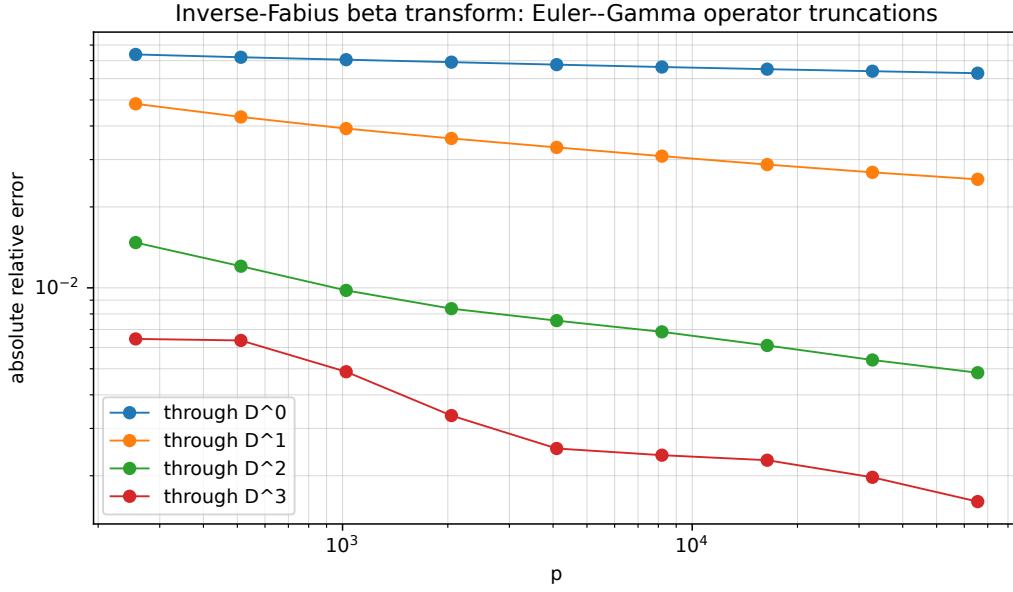


Figure 3: Absolute relative errors of the first four Euler–Gamma operator truncations. The slow scale $\rho^{-1} \asymp (\log p)^{-1/2}$ explains why very large p is needed before successive asymptotic orders separate cleanly.

Conjecture 12.1 (All-orders Gamma–Lambert norm transseries). *There exist one-periodic real-analytic coefficient functions $C_{m,\ell}(\theta)$ such that*

$$\frac{\|\text{up}\|_p^p}{2G_0(p^{-1})} \sim 1 + \sum_{m=1}^{\infty} \rho^{-m} \sum_{\ell=0}^m C_{m,\ell}(\theta) (\log \rho)^\ell, \quad (12.1)$$

with $C_{1,1} = 1/2$ and $C_{1,0} = \kappa_0 - \gamma - \Psi(\theta)$. Every $C_{m,\ell}$ belongs to the differential algebra generated by the inverse saddle jets and the constants $\gamma, \zeta(2), \dots, \zeta(m)$; no additional periodic functions occur.

A proof should combine the formal inverse recursion already in the corpus with a tame remainder theorem for the operator $M_p(\mathcal{D})$. The key analytic task is uniform control of $\mathcal{D}^m G$ on logarithmic windows whose width grows slowly with ρ .

12.2 Nevanlinna parameter of the sinc-square measure

Problem 12.2 (Locate μ_{sp} in the Stieltjes–Wigert Nevanlinna family). Determine the Nevanlinna parameter corresponding to the density (8.5). Equivalently, calculate the Cauchy transform

$$S_{\text{sp}}(z) = \int_0^\infty \frac{d\mu_{\text{sp}}(t)}{z - t} \quad (12.2)$$

relative to the classical entire functions A, B, C, D of the shifted Stieltjes–Wigert moment problem.

The exact q -Pearson law (8.10) supplies unusually rigid extra data. One promising route is to Mellin-transform it, obtaining a functional equation coupling the spectral Mellin transform to the Mellin transform of $\text{sinc}^2(\sqrt{z})$. Another is to use the canonical product of Φ to express S_{sp} as a convergent sum over integer-zero lobes.

Conjecture 12.3 (Sinc-Pearson uniqueness). *Among absolutely continuous representing measures of $m_n = 2^{n(n+1)}$ whose density is locally bounded after multiplication by $\sqrt{z}$, the normalization, moment conditions, and q -Pearson equation (8.10) determine w_{sp} uniquely.*

12.3 Orthogonal complement and spectral zeros

For an indeterminate moment problem, the polynomials are not dense in every representing L^2 space. The spectral measure should possess an orthogonal complement reflecting the integer-zero structure of Φ .

Problem 12.4 (Explicit nonpolynomial orthogonal basis). Construct functions $Q_j(z)$ such that

$$\int_0^\infty z^n Q_j(z) w_{\text{sp}}(z) dz = 0 \quad (n \geq 0)$$

and such that the Q_j span the orthogonal complement of the shifted Stieltjes–Wigert polynomials in $L^2(\mu_{\text{sp}})$. Dyadic rescalings of sinc factors, integer-lobe indicator functions, and q-Bessel functions from the classical theory are the three natural candidate families.

12.4 Confluent q-Richardson phase tomography

The existing corpus develops Gaussian-q Richardson weights for pure power error germs. The norm expansion (12.1) contains powers multiplied by logarithms. Choose a fixed phase ϑ and a dyadic phase-locked sequence

$$\rho_j = M2^j + \delta, \quad \delta + \beta \equiv \vartheta \pmod{1}, \quad p_j = \exp(L\rho_j^2/2). \quad (12.3)$$

Then $h_j = 1/\rho_j$ is an analytic function of 2^{-j} , while the phase is constant. After subtracting the known logarithmic terms, the remaining errors are combinations of $j^\ell 2^{-mj}$.

Conjecture 12.5 (Confluent Gaussian-q filter). *For each truncation order r , there is a stable finite filter whose shift polynomial is a product of repeated factors*

$$\prod_{m=1}^r (E - 2^{-m})^{d_m}, \quad (12.4)$$

which annihilates every error mode $j^\ell 2^{-mj}$ with $0 \leq \ell < d_m$. Its coefficients can be written as derivatives or confluent limits of the Gaussian-q-binomial Lagrange weights

$$w_{r,j}(q) = (-1)^{r-j} \frac{q^{\binom{r-j+1}{2}}}{(q; q)_j (q; q)_{r-j}}. \quad (12.5)$$

Applied to (12.3), this filter recovers $\kappa_0 - \gamma - \Psi(\vartheta)$ with a rigorously controlled error of arbitrarily high order in 2^{-j} .

This would create a genuinely new Richardson pipeline: the data are global derivative norms, the target is a pointwise endpoint periodic function, and the logarithmic multiplicities are handled by confluent q-Lagrange interpolation.

12.5 Base-b and Pascal–Rvachev extensions

The Pascal–Rvachev hierarchy in the corpus replaces the single sinc exponent by binomial multiplicities. The present analysis suggests two extensions.

Problem 12.6 (Norm ladders for higher Rvachev ranks). For each hierarchy level, identify a spatial refinement formula with disjoint or finitely overlapping cells, determine whether normalized derivatives are equimeasurable with a fixed profile, and calculate the resulting derivative weight. If exact equimeasurability fails, seek a finite-state transfer operator for value distributions.

Conjecture 12.7 (Higher-rank lognormal moment geometry). *For the rank- r Pascal–Rvachev profile, the normalized spectral energy moments are q -hyperfactorial sequences whose Hankel determinants factor into products of q -Pochhammer symbols with exponents polynomial in the index. The associated orthogonal polynomials should be multilevel q -binomial deformations of Stieltjes–Wigert polynomials.*

12.6 Formalization roadmap

The roadmap is now partly discharged. The completion labels below refer only to the named declarations in `RvachevDerivativeDistribution.lean`; they do not mark the whole report as formalized. A practical order is:

1. **Completed.** The derivative tiling has a public exact cell coordinate, endpoint/range lemmas, signed and absolute cellwise formulas, and signed midpoint corollary.
2. **Absolute part completed.** The normalized absolute law is an exact Borel pushforward equality and has Banach-valued continuous-test forms. The signed finite-mixture identity is also formal for continuous Banach-valued tests and all n , but its simplified signed Rademacher Borel law remains open.
3. **Still open beyond completed prerequisites.** Use the all- $p \geq 0$ real-power moment theorem, midpoint signs, and existing pointwise bounds to derive `eLpNorm`, general rearrangement-invariant, total-variation, and exact inverse-Fabius level-set corollaries.
4. **Open.** Formalize the Stieltjes substitution and beta-transform identity for the totalized inverse.
5. **Open.** Prove the spectral moments by the existing Fourier differentiation API.
6. **Open.** Formalize the Vandermonde determinant and finite q -binomial orthogonality.
7. **Deferred.** Treat the all-orders asymptotic operator after the inverse derivative-jet bounds have a stable public interface.

13 Conclusion

The dyadic derivative formula contains more information than its usual pointwise reading suggests. Because the cells are disjoint and all have the same affine Jacobian, it is an exact renormalization fixed point for value distributions. The entire derivative tower has one absolute profile, amplified by $2^{n(n+1)/2}$ and signed by Thue–Morse. This immediately produces an exact norm spectrum in every rearrangement-invariant space and makes the inverse Fabius function the level-set law of every derivative.

Two dual transforms then emerge. In physical space, the large- p norms are a beta average of the inverse function. The beta dilation converges to an exponential dilation, whose Mellin symbol is $\Gamma(1 + s)$; Bell polynomials, zeta values, and Bernoulli polynomials are therefore forced into the asymptotic operator. Combining it with the lower-Lambert endpoint coordinate reveals a universal Euler-constant shift and lets global norms perform phase tomography on the periodic saddle correction.

In frequency space, Parseval turns the same derivative scale into the moment sequence $2^{n(n+1)}$. The infinite sinc product thus supplies an explicit, highly nonclassical Stieltjes–Wigert representing measure. Its integer zeros distinguish it from the lognormal representative even though all polynomial moments agree; its q -Pearson law retains the full sinc-square factor. The resulting q -binomial orthogonal polynomials and Gaussian Lagrange rules give a concrete bridge between spectral energy, moment indeterminacy, and the q -calculus already pervasive in the Fabius theory.

The most promising analytic next step is to identify the sinc-square measure inside the Nevanlinna family. On the formal side, the dyadic-cell and normalized absolute equimeasurability core is now complete. The next compact targets are its `eLpNorm`/rearrangement-invariant and inverse-Fabius level-set consequences, followed by the beta-transform layer. The signed Rademacher Borel simplification and every beta/spectral layer in this report remain frontier statements.

A Recursive TeX corpus ledger

The following 68 paths were present under `Analysis/FabiusFunction/docs` at commit 0341bbf91c b867f4510bc8431cf59878208de771. Archived versions were included in the collision audit but interpreted

according to the archive's provenance ledger.

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66	papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
67	papers/arXiv-2211.13570v1/document.tex
68	papers/ubiq15/ubiq15-corrected.tex

B Reproducibility files

The accompanying archive contains:

- Fabius_Derivative_Norm_Spectrum.tex – this source;
- Fabius_Derivative_Norm_Spectrum.pdf – the rendered report;
- numerical_experiments.py – commented reproduction code;
- figures/ – PDF and PNG versions of the three generated figures;
- data/derivative_norm_ratios.csv;
- data/spectral_moment_checks.csv;
- data/beta_operator_checks.csv;
- data/hankel_determinant_checks.csv; and
- data/run_summary.txt.

The default reproduction command is

```
python numerical_experiments.py
```

The grid size, number of sinc factors, derivative order, and output directory are configurable through command-line options documented by

```
python numerical_experiments.py --help
```

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